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FAULT DETECTION OF SUSPENSION HEIGHT SENSORS WITHIN A SEMI-ACTIVE VEHICLE SUSPENSION

Abstract

A control system for a suspension system of a vehicle may include a set of suspension height sensors that may be operably coupled to a set of wheels of the vehicle, a vehicle speed sensor that may be operably coupled to the vehicle; and a controller that may be operably coupled the set of suspension height sensors and the vehicle speed sensor. The controller may be configured to receive the current suspension height data and the vehicle speed data, may determine whether the vehicle may be in a steady-state condition based on the vehicle speed data, responsive to the vehicle being determined to be in the steady-state condition, may perform a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data; and may determine an operation state of the individual suspension height sensor based on the comparison.

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Background/Summary

TECHNICAL FIELD

[0001] Example embodiments generally relate to the operation of a vehicle's control system and, more particularly, relate to a control system for determining an operation state of a suspension height sensor.

BACKGROUND

[0002] Modern vehicles have begun to utilize semi-active suspensions to continuously modify damping levels of a vehicle's suspension system to adjust to varying road conditions. Suspension height sensors are useful for these modern semi-active suspension systems, as they provide accurate and current suspension height data to help control the damping levels. However, detecting and classifying mechanical-specific suspension height sensor faults is presently difficult. Due to the importance of suspension height data in damping adjustment, it would be useful to be able to detect and classify sensor faults quickly to ensure accurate suspension adjustment.

[0003] Thus, it may be desirable to be able to quickly and accurately define and classify suspension height sensor faults in order to properly notify a vehicle operator or service technician.

BRIEF SUMMARY OF SOME EXAMPLES

[0004] In accordance with an example embodiment, a control system for a suspension system of a vehicle may be provided. The control system may include a set of suspension height sensors that may be operably coupled to a set of wheels of the vehicle for measuring current suspension height data, a vehicle speed sensor that may be operably coupled to the vehicle for measuring vehicle speed data; and a controller that may be operably coupled the set of suspension height sensors and the vehicle speed sensor. The set of suspension height sensors may include multiple instances of an individual suspension height sensor that may be operably coupled to corresponding ones of the set of wheels. The controller may be configured to receive the current suspension height data and the vehicle speed data, may determine whether the vehicle may be in a steady-state condition based on the vehicle speed data, responsive to the vehicle being determined to be in the steady-state condition, may perform a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data; and may determine an operation state of the individual suspension height sensor based on the comparison.

[0005] In another example embodiment, a method for determining an operation state in a suspension system element of a vehicle may be provided. The method may include receiving current suspension height data and vehicle speed data, may determine whether the vehicle may be in a steady-state condition based on the vehicle speed data, responsive to the vehicle being determined to be in the steady-state condition, may perform a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data, and may determine the operation state of the individual suspension height sensor based on the comparison. The sensor suite may include a set of suspension height sensors that may be operably coupled to a set of wheels of the vehicle for measuring the current suspension height data and a vehicle speed sensor that may be operably coupled to vehicle for measuring the vehicle speed. The set of suspension height sensors may include multiple instances of individual suspension height sensor that may be operably coupled to corresponding ones of the set of wheels.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0006] Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0007] FIG. **1** illustrates a block diagram of a vehicle control system in accordance with an example embodiment;

[0008] FIG. **2** illustrates a flow chart of an algorithm for detecting an operation state in an individual suspension height sensor in accordance with an example embodiment;

[0009] FIG. **3** illustrates a flow chart for determining if a fault state may be occurring in accordance with an example embodiment;

[0010] FIG. **4** illustrates a flow chart for classifying a fault state in accordance with an example embodiment;

[0011] FIG. **5** illustrates a variable range, a fixed range, and a fixed threshold used in the control system for fault state determination and classification in accordance with an example embodiment; and

[0012] FIG. **6** illustrates a block diagram of a method in accordance with an example embodiment.

DETAILED DESCRIPTION

[0013] Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

[0014] FIG. **1** illustrates a block diagram of a control system **100** according to an example embodiment. The components of the control system **100** may be incorporated into a vehicle **110** (e.g., may be operably coupled to a chassis of the vehicle **110**, various components of the vehicle **110** and/or electronic control systems of the vehicle **110**). In some cases, the chassis may include or be defined by a frame, and the frame may additionally be formed of one or more casted subframes. Of note, although the components of FIG. **1** may be operably coupled to the vehicle **110** and to each other, it should be appreciated that such connection may be either direct or indirect. Moreover, some of the components of the control system **100** may be connected to the vehicle **110** via intermediate connections to other components either of the chassis or of other electronic and/or mechanical systems or components.

[0015] The control system **100** may include a set of wheel assemblies. For example, the set of wheel assemblies may include a front left wheel assembly **120**, a rear left wheel assembly **130**, a front right wheel assembly **140**, and a rear right wheel assembly **150**. The set of wheel assemblies may include multiple instances of an individual suspension height sensor **160** operably coupled to corresponding ones of the set of wheel assemblies. In some cases, the set of wheel assemblies may include various additional suspension system components which may include corresponding shocks, motion sensors for measuring rate of change of velocity, wheel bearings, brakes assemblies, and wheel hubs that are operably coupled to the corresponding ones of the set of wheel assemblies.

[0016] In an example embodiment, the individual suspension height sensor **160** may measure suspension height data, which may include suspension height, vehicle or wheel assembly ride height, wheel assembly position, or various other parameters of set of wheel assemblies. The various other parameters of the individual wheel assembly may be parameters such as velocity or rate of change of velocity that may be derived from position data, which may be measured by the individual suspension height sensor **160**. The individual suspension height sensor **160** may be a rotary suspension height sensor that may utilize a rotary arm to measure suspension height data. In some cases, the individual suspension height sensor **160** may be a linear position sensor which may be operably coupled to a shock of the wheel assembly. In an example embodiment, the linear

position sensor may be a Hall sensor, which may utilize the Hall effect to determine suspension height data of the corresponding ones of the set of wheel assemblies. The linear position sensor may be an inductive sensor, hall sensor, or sensor/any other sensing technology used to determine suspension movement (either position or rate of change of velocity). In some cases, the individual suspension height sensor **160** may be any type of sensor that may measure suspension height data of the corresponding ones of the set of wheel assemblies. In an example embodiment, the individual suspension height sensor **160** may measure voltage measurements as the suspension height data and may correlate the voltage measurements to determine the suspension height data. [0017] Additionally, the control system **100** may further include a controller **200**. While the controller **200** may have many functions, one such function of the controller **200** may be monitoring the status of other components of the vehicle **110**. The components that the controller **200** may monitor the status of may be included in, or not included in, the control system **100**. In this regard, the controller **200** may be responsible for detecting any faults associated with the components of the vehicle **110**. In an example embodiment, the controller **200** may receive information that may be used to determine the status of various components or subassemblies of the vehicle **110** via various sensors that may be operably coupled to the components or subassemblies. Such sensors may be part of a sensor network and sensors of the sensor network may be operably coupled to the controller **200** (and/or the components or subassemblies). In some cases, the controller **200** may communicate the status of certain components in the vehicle **110** to other components in the vehicle **110**, so that appropriate action may be taken to ensure the vehicle **110** may continue operating without any components that may be experiencing a fault. [0018] In some cases, the controller **200** may include a comparison module **210** and a communication module **220**. The comparison module **210** may be primarily responsible for determining if a fault may be occurring within the control system **100**. The communication module **220** may be responsible for communicating information internally within the vehicle **110** or externally to other controllers or devices. The communication module **220** may communicate wirelessly, via a wired connection, or via any other communication protocol, method, or technique that may provide reliable communication between the controller **200** and the target device or destination. In an example embodiment, the communication module **220** may communicate with a user interface **221** of the vehicle **110** or with a service technician **352**. [0019] FIG. 2 illustrates a flow chart for an algorithm **300** for detecting an operation state in an individual suspension height sensor **160** in accordance with an example embodiment. FIG. 3 and FIG. 4 illustrate more detailed flow charts for operations within the algorithm **300** in accordance with an example embodiment. The algorithm **300** may be run by the controller **200**. The algorithm **300** initially may begin with determining a vehicle parameter at operation **310**. In some cases, the vehicle parameter may be vehicle speed measured by a sensor within the control system **100**. In an example embodiment, the vehicle parameter may be records of operator inputs, vehicle rate of change of velocity, or any number of parameters that may be measured by sensors and/or measurements from within the control system **100**. In an example embodiment, more than one vehicle parameter may be determined by operation **310**. [0020] After the vehicle parameter may be determined, the algorithm **300** determines whether the vehicle may be in a steady-state condition based on the vehicle parameter at operation **315**. The steady-state condition may be defined as occurring when the vehicle parameter may be detected as constant for a steady-state determination time period. In some cases, the vehicle **110** may not have a vehicle parameter be constant for an entire period of the steady-state determination time period, in which the steady-state condition may be defined as occurring when the vehicle parameter may be relatively constant (vehicle parameter values within a certain range or percentage of one another) for the steady-state determination time period. For instance, the vehicle parameter may be vehicle speed data. When vehicle speed data may be within a 5 mile per hour range for a steady-state determination time period of 10 seconds, the vehicle **110** may be determined to be in the

steady-state condition. If the vehicle **110** is not determined to be at the steady-state condition, the algorithm **300** may return to or continue to determine the vehicle parameter at operation **310**. [0021] If the vehicle **110** is determined to be at the steady-state condition, the algorithm **300** may proceed to collecting current suspension height data at operation **330**. The current suspension height data may be suspension height data collected from each respective one of the individual suspension height sensor **160** within a suspension height data collection time period. The suspension height data collection time period may overlap with the steady-state determination time period partially, fully, or not at all. In some cases, the suspension height data collection time period may be a brief time period directly after the steady-state determination time period. The suspension height data collection time period may only include a single value, a plurality of values, or an average of a plurality of values of current suspension height data.

[0022] After collection of the current suspension height data, the algorithm **300** may determine an operation state of each individual suspension height sensor **160** and whether each individual suspension height sensor **160** may be in a fault state **342** at operation **330**. If the individual suspension height sensor **160** may be determined to be in the fault state **342**, the fault state **342** may be classified in operation **340**. To determine the operation state of each individual suspension height sensor **160**, the algorithm **300** may compare the current suspension height data to a variable range **410**, a fixed range **420**, and a fixed threshold **430** shown in FIG. 5.

[0023] FIG. 5 illustrates the variable range **410**, the fixed range **420**, and the fixed threshold **430** in accordance with an example embodiment. The variable range **410** may be determined and adjusted based on previous suspension height data. For instance, using previous suspension height data from each individual suspension height sensor **160**, a variable range lower bound **411** and a variable range upper bound **412** of the variable range **410** may be determined. In some cases, the variable range lower bound **411** and the variable range upper bound **412** may be based on upper and lower data points of the previous suspension height data over a previous suspension height data time period. In an example embodiment, the variable range lower bound **411** and the variable range upper bound **412** may be based on deviation (percentage or fixed value) from an average of the previous suspension height data over the previous suspension height data time period or based on any other mathematical property or operation that may be used to adjust previous suspension height data. In some cases, the previous suspension height data time period may be an hour, day, week, month, or year. In an example embodiment, the algorithm **300** may perform trend analysis and compare current suspension height data to the variable range **410** created with the previous suspension height data of a smaller duration one of the suspension height data time periods to the variable range **410** created with the previous suspension height data of a larger duration one of the suspension height data time periods.

[0024] In some cases, the fixed range **420** may be a predetermined range set by a vehicle's manufacturer based on prior testing. The fixed range **420** may be based on the specific vehicle model, vehicle accessories, vehicle size, or any other vehicle feature that may affect suspension performance. In an example embodiment, the fixed range **420** may be always outside the variable range **410**. The fixed range **420** may include a fixed range lower bound **421** and a fixed range upper bound **422**.

[0025] In some cases, the fixed threshold **430** may be a predetermined value set by a vehicle's manufacturer. The fixed threshold **430** may be based on the specific type of individual suspension height sensor **160**, or any other vehicle feature that may affect suspension performance. The fixed threshold **430** may be substantially below a fixed range lower bound **421** to indicate a different fault than the fixed range **420**.

[0026] In an example embodiment, the variable range **410** may be used to determine if there is the fault state **342** present at operation **331**. Responsive to determining that the fault state **342** is presented, the fixed range **420**, and the fixed threshold **430** may be used to classify the fault state **342**. If the current suspension height data is within the variable range **410**, the operation state of the

individual suspension height sensor **160** may be determined to be in a normal state **343**. In some cases, if the current suspension height data may be within the variable range **410**, the current suspension height data may additionally be within the fixed range **420** and above the fixed threshold **430**. If the current suspension height data may be outside the variable range **410** but inside the fixed range **420**, the fault state **342** may be determined to be occurring, as well as the fault state **342** may be further classified as a minor mechanical fault **344** at operation **347**. In some cases, the minor mechanical fault **344** may be indicative of general sensor wear that may not need immediate replacement or correction such as where the sensor is still operable, but is operating in a degraded state that may render its readings less and less accurate over time.

[0027] In an example embodiment, if the current suspension height data may be outside the fixed range **420** but above the fixed threshold **430**, the fault state **342** may be determined to be occurring, as well as the fault state **342** may be further classified as a major mechanical fault **345** at operation **348**. In some cases, the major mechanical fault **345** may indicate a stuck sensor element, broken link arm, bent bracket, jammed sensor mechanism, or any other mechanical function that may render the individual suspension height sensor **160** impractical so that current suspension height data lacks accuracy and correlation to actual suspension height. The major mechanical fault **345** may require prompt replacement or maintenance to return the individual suspension height sensor **160** to the normal state **343**.

[0028] In an example embodiment, if the current suspension height data is below the fixed threshold **430**, the fault state **342** may be determined to be occurring, as well as the fault state **342** may be classified to be an electrical fault **346** at operation **349**. In some cases, the electrical fault **346** may indicate an electrical short within the individual suspension height sensor **160**. In an example embodiment, the fixed threshold **430** may be barely above 0 volts to indicate an electrical short may be present, while still allowing an indication if a major mechanical fault **345** or minor mechanical fault **344** may be present. The fixed threshold may be set by the vehicle's manufacturer as an electrical functioning limits of the individual suspension height sensor **160**.

[0029] In some cases, if the fault state **342** may be determined to not be occurring and the individual suspension height sensor **160** may be determined to be in the normal state **343**, the algorithm **300** may return to operation **310** to determine vehicle parameters.

[0030] In an example embodiment, algorithm **300** may report the fault state **342** in operation **350** upon determining and classifying the fault state **342** in operations **330** and **340**. Depending on the classification of the fault state **342**, the algorithm **300** may report the fault state **342** to different sources. In some cases, if the fault state **342** may be determined as a minor mechanical fault **344**, a report may be delivered to an operator of the vehicle **110** via a display **112** or user interface **221** of the vehicle **110**. In an example embodiment, the report may be an auditory, visual, or haptic indication that may be transmitted by the communication module **220** of the controller **200** to the operator of the vehicle **110**. In some cases, the report may be transmitted by the communication module **220** to a service technician **352**. In an example embodiment, the report may only be sent to the operator if the fault state **342** may be classified as the minor mechanical fault **344**. In some cases, the report may be sent to both the operator and the service technician **352** if the fault state **342** may be classified as the major mechanical fault **345** or the electrical fault **346**. In an example embodiment, even if the fault state **342** may be determined as occurring, a report may be sent to both, one, or neither of the operator and service technician **352**. In some cases, the report sent to the service technician **352** may include vehicle specification and sensor specifications so that a replacement for the individual suspension height sensor **160** or specific component may be pre-ordered.

[0031] In an example embodiment, the algorithm **300** may determine the operation state utilizing only onboard vehicle resources. For example, the operation state may be determined without the use of an external database or external processing. In some cases, an external database of previous suspension height data may be used by the controller **200**. In an example embodiment, the

controller **200** may upload current suspension height data to an external controller. In some cases, the operation state and the fault state **342** may potentially be classified in a single cycle. The single cycle may only collect one set of current suspension height data and may not need iterative learning or machine learning to determine the operation state. In an example embodiment, machine learning or iterative learning may be used by the controller **200** to optimize the algorithm **300**. In some cases, the variable range **410** may be determined via machine learning.

[0032] In an example embodiment, the variable range **410** may only be one of a plurality of variable ranges. Each variable range **410** of the plurality of variable ranges may be determined based on different ranges of the previous suspension height data time period. For example, the plurality of variable ranges may include a daily variable range, a weekly variable range, a monthly variable range, and a lifetime variable range. The plurality of variable ranges may be based on any number of the previous suspension height data time periods. In some cases, the algorithm **300** may utilize the plurality of variable ranges to perform trend analysis on the individual suspension height sensor **160**. Trend analysis may be performed by comparing the current suspension height data to the plurality of variable ranges, as well as comparing the plurality of variable ranges to one another. In an example embodiment, the operation state may be determined via trend analysis.

[0033] In some cases, the algorithm **300** may include a plurality of fixed ranges. The plurality of fixed ranges may be set by the vehicle's manufacture to define specific mechanical functioning limits of the vehicle **110**. The plurality of fixed ranges may be used in trend analysis similar to the plurality of variable ranges.

[0034] In an example embodiment, the algorithm **300** may include a plurality of fixed thresholds. The plurality of fixed thresholds may be set by the vehicle's manufacture to define electrical functioning limits of the vehicle **110**. The electrical functioning limits may allow the determination if the electrical fault **346** may be a complete short or some other electrical issue. The plurality of fixed thresholds may be used in trend analysis similar to the plurality of variable ranges and the plurality of fixed ranges.

[0035] In some cases, trend analysis may be performed using the plurality of variable ranges, the plurality of fixed ranges, and the plurality of fixed thresholds together. In an example embodiment, the algorithm **300** may be run continuously and automatically by the controller **200**.

[0036] FIG. **6** illustrates a block diagram of a method for determining an operation state of the individual suspension height sensor in accordance with an example embodiment. The method may include receiving current suspension height data and vehicle speed data at operation **600**. The method may further include determining whether the vehicle maybe in a steady-state condition based on the vehicle speed data in operation **610**. The method also may include, responsive to if in steady-state condition may be present, perform a comparison of the current suspension height data to the variable range, the fixed range, and the fixed threshold of suspension height data at operation **620**. Finally, the method may include determining the operation state of the individual suspension height sensor based on the comparison at operation **630**.

[0037] A control system for a suspension system of a vehicle may therefore be provided. The control system may include a set of suspension height sensors that may be operably coupled to a set of wheels of the vehicle for measuring current suspension height data, a vehicle speed sensor that may be operably coupled to the vehicle for measuring vehicle speed data; and a controller that may be operably coupled the set of suspension height sensors and the vehicle speed sensor. The set of suspension height sensors may include multiple instances of an individual suspension height sensor that may be operably coupled to corresponding ones of the set of wheels. The controller may be configured to receive the current suspension height data and the vehicle speed data, may determine whether the vehicle may be in a steady-state condition based on the vehicle speed data, responsive to the vehicle being may being determined to be in the steady-state condition, may perform a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data; and may determine an operation state of the individual

suspension height sensor based on the comparison.

[0038] The control system for a suspension system of a vehicle of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the suspension assembly. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the comparison may classify the operation state as either a normal state or a fault state, and responsive to determining the operation state of the individual height sensor may be the fault state, the controller may classify a cause of the fault state based on the comparison. The cause of the fault state may be classified as either a mechanical fault or an electrical fault. In an example embodiment, responsive to the current suspension height data being within the variable range, the operation state may be classified as the normal state. In some cases, responsive to the current suspension height data being outside the variable range but within the fixed range, the operation state may be classified to be the fault state and further classified to be a minor mechanical fault. In an example embodiment, responsive to the current suspension height data being outside the fixed range but above the fixed threshold, the operation state may be classified to be the fault state and further classified to be a major mechanical fault. In some cases, responsive to the current suspension height data being below the fixed threshold, the operation state may be classified to be the fault state and further classified to be an electrical fault. In an example embodiment, the current suspension height data and the suspension height data may be voltage measurements. In some cases, the variable range may be within the fixed range, and the fixed threshold may be below a lower limit of the fixed range. In an example embodiment, the variable range may be adjusted based on previously measured suspension height data of the vehicle. In some cases, the fixed range may be predetermined based on vehicle classification or vehicle model prior to any operation of the vehicle. In an example embodiment, responsive to determining the operation state may be the fault state, the controller may report a cause of the fault state to an operator of the vehicle via a user interface or to a service technician depending on the cause of the fault state. In some cases, the controller may determine the operation state of the individual suspension height sensor via only onboard vehicle resources. In an example embodiment, the fault state of the individual suspension height sensor may be determined in a single cycle.

[0039] A method for determining an operation state in a suspension system element of a vehicle may be provided. The method may include receiving current suspension height data and vehicle speed data, may determine whether the vehicle may be in a steady-state condition based on the vehicle speed data, responsive to the vehicle being determined to be in the steady-state condition, may perform a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data, and may determine the operation state of the individual suspension height sensor based on the comparison. The sensor suite may include a set of suspension height sensors that may be operably coupled to a set of wheels of the vehicle for measuring the current suspension height data and a vehicle speed sensor that may be operably coupled to vehicle for measuring the vehicle speed. The set of suspension height sensors may include multiple instances of individual suspension height sensor that may be operably coupled to corresponding ones of the set of wheels.

[0040] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or

functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

- 1.** A control system for a suspension system of a vehicle, the control system comprising: a set of suspension height sensors operably coupled to a set of wheels of the vehicle for measuring current suspension height data; a vehicle speed sensor operably coupled to the vehicle for measuring vehicle speed data; and a controller operably coupled the set of suspension height sensors and the vehicle speed sensor, wherein the set of suspension height sensors comprises multiple instances of an individual suspension height sensor operably coupled to corresponding ones of the set of wheels, wherein the controller is configured to: receive the current suspension height data and the vehicle speed data; determine whether the vehicle is in a steady-state condition based on the vehicle speed data; responsive to the vehicle being determined to be in the steady-state condition, perform a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data; and determine an operation state of the individual suspension height sensor based on the comparison.
- 2.** The control system of claim 1, wherein the comparison classifies the operation state as either a normal state or a fault state, wherein responsive to determining the operation state of the individual height sensor is the fault state, the controller further classifies a cause of the fault state based on the comparison, and wherein the cause of the fault state is classified as either a mechanical fault or an electrical fault.
- 3.** The control system of claim 2, wherein responsive to the current suspension height data being within the variable range, the operation state is classified as the normal state.
- 4.** The control system of claim 2, wherein responsive to the current suspension height data being outside the variable range but within the fixed range, the operation state is classified to be the fault state and further classified to be a minor mechanical fault.
- 5.** The control system of claim 2, wherein responsive to the current suspension height data being outside the fixed range but above the fixed threshold, the operation state is classified to be the fault state and further classified to be a major mechanical fault.
- 6.** The control system of claim 2, wherein responsive to the current suspension height data being below the fixed threshold, the operation state is classified to be the fault state and further classified to be an electrical fault.
- 7.** The control system of claim 2, wherein responsive to determining the operation state is the fault state, the controller reports a cause of the fault state to an operator of the vehicle via a user interface or to a service technician depending on the cause of the fault state.
- 8.** The control system of claim 1, wherein the variable range is within the fixed range and the fixed threshold is below a lower limit of the fixed range.
- 9.** The control system of claim 1, wherein the variable range is adjusted based on previously measured suspension height data of the vehicle.
- 10.** The control system of claim 1, wherein the fixed range is predetermined based on vehicle classification or vehicle model prior to any operation of the vehicle.

- 11.** The control system of claim 1, wherein the current suspension height data and the suspension height data are voltage measurements.
 - 12.** The control system of claim 1, wherein the controller determines the operation state of the individual suspension height sensor via only onboard vehicle resources.
 - 13.** The control system of claim 1, wherein the operation state of the individual suspension height sensor is determined in a single cycle.
 - 14.** A method for determining an operation state in a suspension system element of a vehicle, the method comprising: receiving current suspension height data and vehicle speed data, determining whether the vehicle is in a steady-state condition based on the vehicle speed data; responsive to the vehicle being determined to be in the steady-state condition, performing a comparison of the current suspension height data to a variable range, a fixed range, and a fixed threshold of suspension height data; and determining the operation state of the individual suspension height sensor based on the comparison, wherein a set of suspension height sensors operably coupled to a set of wheels of the vehicle for measuring the current suspension height data and a vehicle speed sensor operably coupled to vehicle for measuring the vehicle speed, and wherein the set of suspension height sensors comprises multiple instances of individual suspension height sensor operably coupled to corresponding ones of the set of wheels.
 - 15.** The method of claim 14, wherein the comparison classifies the operation state as either a normal state or a fault state, wherein responsive to determining the operation state of the individual height sensor is the fault state, the method further includes classifying a cause of the fault state based on the comparison, and wherein the cause of the fault state is classified as either a mechanical fault or an electrical fault.
 - 16.** The method of claim 15, wherein responsive to the current suspension height data being within the variable range, the operation state is classified as the normal state.
 - 17.** The method of claim 15, wherein responsive to the current suspension height data being outside the variable range but within the fixed range, the operation state is classified to be the fault state and further classified to be a minor mechanical fault.
 - 18.** The method of claim 15, wherein responsive to the current suspension height data being outside the fixed range but above the fixed threshold, the operation state is classified to be the fault state and further classified to be a major mechanical fault.
 - 19.** The method of claim 15, wherein responsive to the current suspension height data being below the fixed threshold, the operation state is classified to be the fault state and further classified to be an electrical fault.
 - 20.** The method of claim 15, wherein responsive to determining the operation state is the fault state, the method further includes reporting a cause of the fault state to an operator of the vehicle via a user interface or to a service technician depending on the cause of the fault state.
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